Distortion Resistant Venous Reservoir Blood Volume Sensor: Applied Computer Vision in Cardiac Surgery

Lawrence Technological University
Be curious. Make magic.

Orrum CLINICAL ANALYTICS

Alec Lipinski[1], Ryan Kaddis[2], Sean Murtha[3], Robb Johnson[3], Chan-Jin Chung[2], Hao Jiang[1]

[1]Dept. of Biomedical Engineering, CoE; [2]Dept. of Math and Computer Science, CoAS; [3]Orrum Clinical Analytics, Inc.

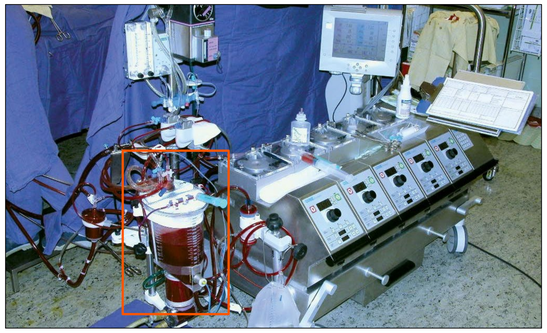
Emails: cchung@ltu.edu; hjiang@ltu.edu

Introduction

Clinical Context & Problem

- **Cardiopulmonary Bypass (CPB)** in cardiac surgery requires venous blood storage/monitoring
- Blood volume must be monitored to prevent serious injury
  - Air embolism, cerebral injury (volume too high/low)
  - Organ damage (rapid volume changes)
- Perfusionists **manually** read and **intermittently** chart **venous reservoir blood volume (VRBV)**
- Automated VRBV acquisition is not standard clinical practice, despite its potential to predict key physiological parameters

Fig. 1: The Heart-Lung Machine & Venous Reservoir



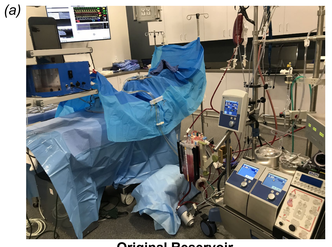
The venous reservoir is one of the only components in the CPB circuit lacking automatic, constant data acquisition.

Previous Solutions

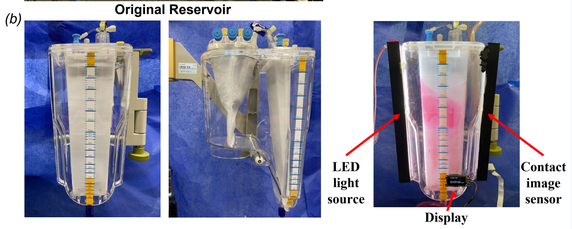
- Optical Sensor – Scans blood level using Contact Image Sensor
  - Accurate, precise, non-intrusive
  - Blocks reservoir visibility
- Gravimetric sensor – Measures volume via weight
  - Accurate, precise, un-obstructive
  - Requires zero-point calibration, affected by movement

Fig. 2: VRBV Sensor Based on a Contact Image Sensing[1]

(a)



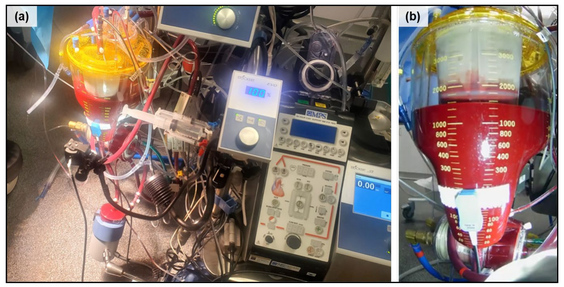
(b)



(a) Experimental setup for testing the TERUMO reservoir with the contact image sensor.
(b) Comparison of the original TERUMO reservoir and one equipped with the contact image blood volume sensor.

Off-Reservoir Computer Vision Sensor

Fig. 3: VRBV Sensor Based on Computer Vision[2]



Our previous computer vision system, tested in a Simulated CPB environment at the Cardiovascular Perfusion SIMLAB, Orrum Clinical Analytics Inc.

- **Off-Reservoir Computer Vision System**
  - Accurate, precise, un-obstructive
  - Did not correct for image distortion

Objectives

Venous Reservoir Blood Volume (VRBV) Sensor

- Mimics perfusionists' blood volume interpretation
- Convenient for use with clinical CPB
- Accurate (Close to True Value) and precise (Consistent)
- Non-intrusive: no contact with blood
- Non-obstructive: does not block perfusionists' view of the reservoir
- Versatile: compatible with all major brands and models of reservoir
- **Is compatible with image distortion resulting from:**
  - Mounting Position (Close Viewing Distance, Steep Angle)
  - Camera's Intrinsic Characteristics (Wide Angle Lens)

Materials & Methods

Equipment

- Sensor System:
  - HD High Dynamic Range Webcam – Arducam IMX291
  - LED Lamp with Diffusing Layer
  - Computer – Lenovo ThinkPad P1 Gen 5 Laptop, equipped with NVIDIA RTX A2000 GPU
- Custom Mounting System Displaying Laser Engraved ArUco Markers
- Venous Reservoirs – SORIN INSPIRE HVR & TERUMO CAPIOX NX19
- Test Liquid – Water, Cornstarch, Red Food Dye

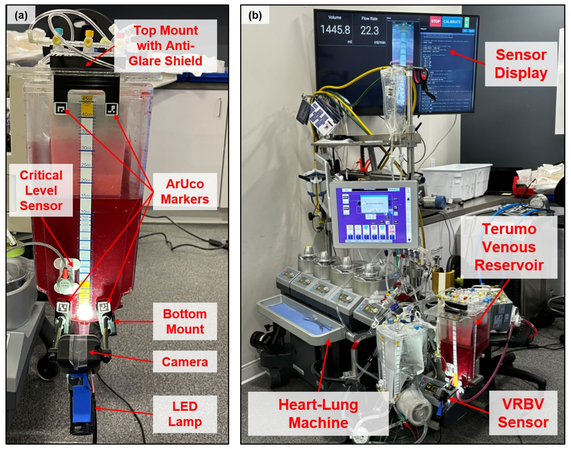
Data Acquisition Protocols

1. Fill reservoir with test liquid
2. Liquid level set to major grid line – e.g. 3000 mL
3. Grid Line = "Ground Truth" Volume
4. Compare "Ground Truth" Vs. Sensor Reading
5. Accuracy: **Mean Absolute Percentage Error (MAPE)**
6. Precision: **Standard Deviation**

Experimental Results

- Tested sensor response to simulated blood under high perspective distortion resulting from **on-reservoir** mounting system
- Recorded MAPE and standard deviation with critical-level sensor installed

Fig. 4: Sensor Configuration & Setup within Simulated Clinical CPB Environment

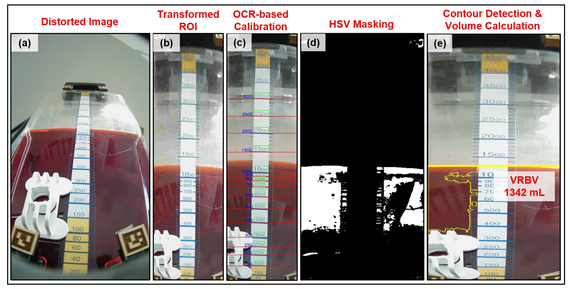


(a) VRBV sensor configured in a rigid mount on the Terumo Capiox NX19 Reservoir. (b) Image of the sensor installed in Orrum Clinical Analytics SIMLAB.

How Does it Work?

1. Image Transformation Removes Perspective Distortion
   - ArUco Markers Provide 4 coordinates corresponding to the new image
2. Hue, Saturation, Value (HSV) Masking detects blood volume
3. Optical Character Recognition (OCR) locates volume labels
4. Contour detection locates top edge of the blood-colored region
5. Venous Reservoir Blood Volume is Interpolated/Extrapolated from known label positions and fluid level

Fig. 5: Distortion-Resistant Algorithm for VRBV Detection



(a) A distorted frame acquired from video stream, (b) processed with perspective transformation. (c) One-time label registration based on OCR detection. (d) HSV masking and (e) contour detection used in extracting the blood level.

Table 1: Sensor Performance using Distortion Correction on the Terumo Reservoir

Ground Truth (mL)	Average Reading (mL)	MAPE (%)	Standard Deviation (mL)
3000	2991.19	0.29	3.398
2000	1959.65	2.59	196.358
1000	1010.21	1.15	12.044
800	802.10	0.59	5.754
600	595.74	0.84	4.048
400	374.58	6.36	6.315
300	310.45	3.66	9.125
200	195.07	6.15	14.074
150	161.07	7.38	4.851

Test Results from Simulated CPB Environment

- Average MAPE across the test volumes was 3.22%
  - Results are consistent with the off-reservoir sensor
- Accurate and precise volume readings
- The system meets clinical CPB requirements without hindering perfusionists' operation

Conclusions & Future Work

Integrating distortion correction into our computer vision sensor significantly enhances the sensor's versatility, enabling accurate venous reservoir blood volume tracking across variable mounting angles and viewpoints.

Future Work:

- Improve Environmental Robustness by phasing out:
  - OCR for label detection: Computationally expensive and requires high resolution
  - Color-based blood detection: prone to failure from shadows and surface glare
  - Both methods exhibit high sensitivity to variable clinical lighting conditions

Acknowledgements

- Funded by Orrum Clinical Analytics Inc.
- Special thanks to Andrew Ulaszek for assisting 3D printing and ordering materials.

Journal & Patent Publications

[1] T. Foley, A. F.-R. i Sabala, M. Fockler, Z. Alzayer, S. Murtha, and H. Jiang, "Continuous contactless measurement of blood volume inside venous reservoirs for cardiopulmonary bypass," **IEEE Sensors Journal**, vol. 23, no. 13, pp. 14 882– 14 890, 2023.

[2] R. Kaddis, C.J. Chung, S. Murtha and H. Jiang, "A Non-intrusive, Non-obstructive, Versatile Venous Reservoir Blood Volume Sensor Based on Computer Vision for Clinical Cardiopulmonary Bypass," **IEEE Sensors Letters**, vol. 9, no. 5, pp. 1-4, May 2025, Art no. 6004604

[3] S. Murtha, H. Jiang, T. Foley, A. F.-R. i Sabala, M. Fockler, Z. Alzayer, and A. Wayne, "Blood volume sensor system," **PCT Patent Application**, p. WO2023/107591A1, 2023. [Patent pending]

[4] Sean Murtha, Ryan Kaddis, Chan-Jin Chung, Hao Jiang., and Alec Lipinski. "Venous Reservoir Blood Volume Sensor with Computer Vision", **U.S. Patent Application No. PCT/US25/51197**, filed in Oct, 2025. [Patent pending]